

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Complete the following:

1 / 1 point

There are 10 boxes and 11 items. At least one box contains...items.

☒ At least two

☐ Exactly two

☐ At most two

☐ Two

✔ Nice work

This is correct. We have 11 pigeons and 10 pigeonholes. Using the Pigeonhole Principle, there are at least $\lceil 11/10 \rceil = 2$ pigeons (items) in at least one pigeonhole (box).
2. Complete the statement:

1 / 1 point

If there are N objects to be placed in k boxes, there is at least one box containing at least...objects.

☐ $N - k$

☐ $k.N$

☐ $\lfloor N/k \rfloor$

☒ $\lceil N/k \rceil$

✔ Nice work

This is correct. It is the Pigeonhole Principle.
3. How many cards (minimum) from a standard deck of 52 cards must be selected to guarantee that at least 10 cards are from the same suit?

1 / 1 point

☒ 37

☐ 10

☐ 11

☐ 40

✔ Nice work

This is correct. There are four pigeonholes, each representing a suit. If each pigeonhole contains nine cards, in total we have $4 \cdot 9 = 36$ cards. So, selecting the 37th card would result in one of those pigeonholes having at least 10 cards (from the same suit).
4. How many times (minimum) should a die be rolled to guarantee that one number is seen at least 10 times?

1 / 1 point

☒ 55

☐ 11

☐ 60

☐ 10

✔ Nice work

This is correct. There are six pigeonholes representing a number on die. If each number is seen nine times, the die should be rolled 54 times so that the 55th time guarantees that one number is seen at least three times.
5. A bag contains six blue balls and five red balls. How many balls (minimum) must be selected to guarantee that at least three balls are the same colour?

1 / 1 point

☒ 5

☐ Four

☐ Six

☐ Three

✔ Nice work

This is correct. If we select two blue balls and two red balls, then the next one guarantees that at least three balls are the same colour (it does not matter if it is blue or red). So, the answer is five.
6. A bag contains five blue balls, four red balls and four green balls. How many balls (minimum) must be selected to guarantee that at least four balls are the same colour?

1 / 1 point

☐ Five

☒ 10

☐ Four

☐ 12

✔ Nice work

This is correct. If we select three blue balls, three red balls and three green balls then the next one guarantees that at least four balls are the same colour (it does not matter if it is blue, red or green). So, the answer is 10.